

Shower Device

Happy Campers

Element J: Documentation of Evaluation

Dormitory residents and dormitory guests of our boarding school need a device to place their shower products. This is because they would want their shower products to be easily accessible in the communal shower. They would also like to do this without having a personal shower caddy and without setting it on the floor because it is unsanitary. The main stakeholders are the dormitory residents. They need a solution because the showers in the dormitory were not built with a solution to this problem and dormitory residents are not permitted to leave their shower products in the shower area. The only place to put a device in the dormitory showers are on a relatively narrow divider wall of less than an inch wide.

The three experts the team works with: Z, E; Mr. S, M; Ms. Z, C; are qualified to review the team's final project because of the followings: Zimath is an extremely knowledgeable person in the field of 3D printing, he worked with a various types of filaments, and have made outstanding works with the help of 3D printing technology; Scullin is the physics and chemistry teacher in the school, which made him the expert in structure analysis of the project; Zito is the science and engineering teacher, and the director of applied engineering program of the school. She is extremely familiar with the engineering design process as well as the project.

Critiques given to us are that the team should remove the need to create a hole through the shower divider wall, and instead have the weight and balance of the device be solved by going over the shower divider wall over the divider wall instead of through it, according to one of the team's expert, Zito.

Another feedback provided to the team is in consideration of the small printing platform the team can access. Therefore the team was recommended to try to print the design vertically.

Based on the feedback from the exports, the team attempted to print the design vertically, but eventually printed it in two pieces because of the lack of z-axis printability and z-axis strength. Additionally, the team used nylon ropes to secure the design on the divider as it carries a few essential physical properties including hydrophobic and high mechanical strength.